

Pious Deo

+31-616924521 • Alkmaar, Netherlands • piousdeo36@gmail.com •  [Pious Deo](#)

Mechanical Design Engineer

Results-driven Mechanical Design Engineer with 4+ years of experience, including 3+ years in high-tech industry. Skilled in the full product development lifecycle i.e. from requirements engineering and concept development through detailed design, risk analysis (FMEA), procurement, assembly & integration, and SAT. Proficient in SolidWorks with hands-on experience in DFM and tolerance analysis (GD&T). Seeking a mid-level design engineering role in the mechanical or machine building industry.

WORK EXPERIENCE

E-Magy B.V. – *Broek op Langedijk, Netherlands*

Jun. 2023-Present

Mechanical Engineer

- Delivered 15+ end-to-end R&D projects to scale a special purpose metal alloy casting machine from finite batch to continuous production owning all phases from requirements definition through site acceptance testing (SAT) ensuring designs met applicable NEN/EN and CE Marking requirements throughout.
- Engineered conceptual and detailed designs in SolidWorks including sheet metal components incorporating mechanical & thermal analysis, and executed FMEA on 30+ sub-systems to convert process-specific failure modes into design changes before hardware build, maximizing machine uptime and reliability.
- Produced GD&T-controlled engineering drawings for 100+ custom components, BoM and managed full supplier procurement cycle from RFQ through delivery, ensuring dimensional quality and schedule adherence for custom casting machine components.
- Commissioned casting machine builds through mechanical assembly and structured SAT execution, validating continuous-run performance against process targets in collaboration with cross-functional teams.

MARU Systems – *Zuidoostbeemster, Netherlands*

Jan. 2022-May 2023

Team Lead Mechanical Engineer

Jan. 2023-May 2023

- Led a team of 4 engineers across mechanical improvement projects for production equipment, owning scope definition, planning, and task delegation end-to-end.
- Owned technical documentation including FMEA and risk analysis, and gained hands-on exposure to CE Marking certification and Eurocodes compliance.

Mechanical Engineer

Jan. 2022-Jan. 2023

- Developed innovative concepts for hybrid/off-grid power solutions using Inventor 2022, supported by FEA structural simulations in ANSYS to validate steel frame performance.
- Delivered full product hardware documentation such as BoM, 2D manufacturing drawings and supported prototype builds in close collaboration with production and suppliers.

TECHNICAL SKILLS

- CAD – SolidWorks, Inventor, CATIA
- FEA – ANSYS, ABAQUS
- Manufacturing & Fabrication – Sheet metal design, DFM, DFA, tolerance analysis (GD&T)
- Standards & Compliance – CE Marking & NEN/EN standards.
- PDM/PLM – SolidWorks PDM

PROFESSIONAL CERTIFICATIONS

- Manufacturing Process selection and Design for Manufacturing (DFM) issued by Udemy
- ATEX Awareness Training by KWA Bedrijfsadviseurs
- FMEA Specialist Certification issued by AIGPE/Udemy
- Safety for Operational Supervisors (VCA VOL)
- GD&T Fundamentals issued by GD&T Basics

EDUCATION

University of Twente - Enschede, *Netherlands*

MSc. Mechanical Engineering

Aug. 2019 - Aug. 2021

Thesis – Towards VARTM Manufacturing Process Optimization: Analysis of Temperature and Cure

FEA-based optimization of a VARTM composite manufacturing process which included development of a transient heat transfer model in ABAQUS which was validated against experimental cure and temperature data.

Savitribai Phule Pune University - Pune, India

BSc. Mechanical Engineering

Jun. 2013 - Jun. 2017

LANGUAGE COMPETENCIES

- English – Professional working proficiency (C1)
- Dutch – Basic (A2)
- Hindi/Marathi – Native speaker